Automatic Rebar Detailing System

Engineering Design Report

Project Details

Project Name: Slab Regression Case
Designer Name: N/A
Date: N/A
Remarks: N/A

INPUTS

Length: 5 m
Width: 3 m
Slab Thickness: 150 mm
Support/Wall Thickness: 150 mm
Clear Cover: 25 mm
Floor Finish Load: 1.5 kN/m²
Live Load: 3 kN/m²
Concrete Grade: M35
Steel Grade: Fe500
Support Type: Simply Supported
Unit Weight of Concrete: 25 kN/m³
Preferred Bar Diameter: 12 mm

LOAD CALCULATION

Slab thickness: 150 mm
Self weight: 3.75 kN/m²
Floor finish: 1.5 kN/m²
Dead load: 5.25 kN/m²
Live load: 3 kN/m²
Service load: 8.25 kN/m²
Ultimate load: 12.375 kN/m²

GEOMETRY

Clear span: 2.85 m
Effective short span: 3 m
Effective long span: 5 m
Aspect ratio ly/lx: 1.667
Slab classification: Two Way Slab
Effective span method: IS 456:2000 Clause 22.2: effective span = clear span + support width; for the provided span input, clear span = total span - support width and effective span = total span. Short and long directions calculated independently.

TWO-WAY MOMENT DESIGN

Coefficient short: 0.0497
Coefficient long: 0.0337
Mu short: 5.535 kN·m/m
Mu long: 3.753 kN·m/m
Moment equation: Mx = alphaX x w x lx^2; My = alphaY x w x lx^2 (IS 456 Table 26 coefficient convention).
Short span used for both moments, lx: 3 m
Coefficient source: IS 456:2000 Table 26 with linear interpolation between m = 1.6 and 1.7.

FLEXURAL DESIGN

SHORT DIRECTION
b: 1000 mm
d: 121 mm
Mu: 5.535 kN·m/m
Ast required: 180 mm²

Ast minimum: 180 mm²
Ast provided: 182.78 mm²
Bar arrangement: 8 mm @ 275 mm c/c
xu: 6.31 mm
xu,max: 55.66 mm
Moment capacity: 9.41 kN·m/m
Status: PASS

LONG DIRECTION
b: 1000 mm
d: 113 mm
Mu: 3.753 kN·m/m
Ast required: 180 mm²
Ast minimum: 180 mm²
Ast provided: 182.78 mm²
Bar arrangement: 8 mm @ 275 mm c/c
xu: 6.31 mm
xu,max: 51.98 mm
Moment capacity: 8.77 kN·m/m
Status: PASS

SHEAR

Critical section: At d = 121 mm from the support face; one-metre strip reaction = 18.563 kN/m.
Vu: 17.065 kN/m
tau_v: 0.141 N/mm²
tau_c: 0.3008 N/mm²
tau_c,max: 3.7 N/mm²
Tension steel percentage for Table 19: 0.1511 %
Concrete shear capacity: 36.402 kN/m
Reinforcement: Not required / not applicable because concrete shear capacity is adequate.
Status: PASS

DEFLECTION

Governing direction: Long direction
Governing span: 5 m
Governing effective depth: 113 mm
Basic L/d: 20
Tension steel percentage: 0.1618 %
Modification factor: 1.0667
Actual L/d: 44.25
Permissible L/d: 21.33
Method: IS 456:2000 Clause 23.2.1 basic span/depth ratio multiplied by the Fig. 4 modification factor.
Source: IS 456:2000 Clause 23.2.1 and Fig. 4; configurable digitized design points for modification factor interpolation.
Interpolation: Linear interpolation in tension steel percentage and service steel stress; values are clamped to the configured chart bounds.
Status: FAIL

DETAILING

Short reinforcement: 8 mm @ 275 mm c/c
Long reinforcement: 8 mm @ 275 mm c/c
Short spacing: 275 mm
Long spacing: 275 mm
Short maximum spacing: 300 mm
Long maximum spacing: 300 mm
Short spacing status: PASS
Long spacing status: PASS
Short clear spacing: 267 mm
Long clear spacing: 267 mm
Minimum clear spacing: 25 mm

Clear-spacing rule: IS 456:2000 Clause 26.3.2: adopted minimum clear spacing is 25 mm for this design input set; the larger of bar diameter and 5 mm above nominal maximum aggregate size governs when aggregate size is supplied.
Maximum-spacing rule: IS 456:2000 Clause 26.3.3: main reinforcement spacing shall not exceed 3d or 300 mm, whichever is less. Both directions of a two-way slab are treated as main flexural reinforcement.
Short spacing check: PASS
Long spacing check: PASS
Cover: 25 mm
Corner restraint: FREE_TO_LIFT
Corner torsion reinforcement: Not required under the explicit FREE_TO_LIFT assumption.
Corner torsion status: Not required

FINAL CHECKS

Effective span validity: PASS | actual 5 | limit 0 | Effective span is finite and valid.
Slab classification: PASS | actual 1.667 | limit 2 | Aspect ratio = 1.667 -> Two Way Slab.
Short-direction flexural capacity: PASS | actual 9.41 | limit 5.535 | MuProvided 9.41 >= required 5.535 and xu <= xu,max.
Long-direction flexural capacity: PASS | actual 8.77 | limit 3.753 | MuProvided 8.77 >= required 3.753 and xu <= xu,max.
Short required reinforcement: PASS | actual 182.78 | limit 180 | AstProvided 182.78 >= AstRequired 180.
Long required reinforcement: PASS | actual 182.78 | limit 180 | AstProvided 182.78 >= AstRequired 180.
Short minimum reinforcement: PASS | actual 182.78 | limit 180 | AstProvided 182.78 >= AstMin 180.
Long minimum reinforcement: PASS | actual 182.78 | limit 180 | AstProvided 182.78 >= AstMin 180.
Short maximum spacing: PASS | actual 275 | limit 300 | Provided spacing 275 <= max 300.
Long maximum spacing: PASS | actual 275 | limit 300 | Provided spacing 275 <= max 300.
Short clear spacing: PASS | actual 267 | limit 25 | Clear spacing 267 >= minimum 25 mm.
Long clear spacing: PASS | actual 267 | limit 25 | Clear spacing 267 >= minimum 25 mm.
Shear check: PASS | actual 0.141 | limit 0.3008 | Shear stress is within concrete shear capacity.
Deflection control: FAIL | actual 44.25 | limit 21.33 | Actual L/d exceeds the permissible modified L/d; slab thickness/depth is insufficient for deflection control.
Cover and detailing: PASS | actual 25 | limit 150 | Nominal cover 25 mm is valid, slab thickness 150 mm.
Corner torsion restraint: PASS | actual 0 | limit 0 | Corners are explicitly configured as free to lift; corner torsion reinforcement is not required under this assumption.
Final status: NOT APPROVED